Explainable Semantic Textual Similarity via Dissimilar Span Detection

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Dissimilar Span Detection: Given two sequences of word or sub-word tokens $X = (x_1, x_2, ..., x_n)$ and $Y = (y_1, y_2, ..., y_m)$, let $s_j := ((x_a, x_{a+c}), (y_b, y_{b+d}))$ be a token span pair coming from X and Y, with $1 \le a < n$, $1 \le b < m$, $1 \le c \le n - a$ and $1 \le d \le m - b$, the goal is to find all non-overlapping span pairs s_j that, having a common semantic function, differ in meaning.

Sentence pair	STS
On April 15, 2010, the Colorado Court of Appeals dismissed. On March 15, 2012, the Oregon Court of Appeals dismissed.	0.74
She went on to win a silver medal in the 2024 Olympics. She went on to win first place in the 2024 Olympics.	0.91
They are all over the legal age, so such behavior is legal. They are all over 18, so such behavior is not illegal.	0.88

→ **Dissimilar Span Detection** via LLMs, Sentence Transformers, etc. →

On [April 15, 2010], the [Colorado Court of Appeals] dismissed.
On [March 15, 2012], the [Oregon Court of Appeals] dismissed.

She went on to win a [silver medal] in the 2024 Olympics.
She went on to win [first place] in the 2024 Olympics.

They are all over the legal age, so such behavior is legal.
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Downstream Tasks

- **Fake News Detection**: Identify key mis-matching information between sources.
- **Text Clustering**: Avoid clustering texts with high STS but differing information.
- **Annotation Error Detection**: Detect annotation errors in STS-related datasets.
- **Critical Error Detection**: In a multilingual setting, detect errors when evaluating MT systems.
- **Parallel Corpus Mining**: In a multilingual setting, avoid matching pairs with key differing information.
- Etc.

1. Span Similarity Dataset (SSD)

New dataset (1,000 samples) crafted specifically for the task of DSD:

Sentence 1	Sentence 2	Span Similarity	Sentence Similarity
It was {{restored}} in the {{1980s}}.	It was {{destroyed}} in the {{eighties}}.	0,1	0
Thank you for {{wasting my money}}.	Thank you for {{misusing my funds}}.	1	1
There are depots at {{Quilpie}} and {{Roma}}.	There are depots at {{Brisbane}} and {{Sydney}}.	0,0	0

Table 1: Examples of shorter sentences from the SSD. Spans are denoted using double curly braces, and then annotated respectively with a 0, in case the span pair differs in meaning, or 1, if they are equivalent. We also annotate the sentence similarity of the pair as a whole.

Created in a semi-automatic way in two steps: (1) We let an LLM replace the spans, and (2) we manually review them and annotate the similarity. To ensure the correctness of the spans, six annotators manually annotate spans in 100 samples, achieving good inter-annotator and annotator-dataset agreement ($\kappa \in [0.61, 0.91]$).

3. Results

Method	Model Size	SSD F1-Global	SSD F1-NoDiff	SSD F1-Diff	SSD Time	SemEval-2016 F1-Diff	SemEval-2016 Time
LIME `all-mpnet-base-v2`	109M	0.463	0.782	0.223	1981.81	0.109	199.89
SHAP `all-MiniLM-L6-v2`	22.7M	0.366	0.434	0.131	44.47	0.306	4.52
Embedding `all-mpnet-base-v2`	109M	0.469	0.529	0.423	11.28	0.352	1.13
Embedding `text-embedding-004`	Billions	0.547	0.666	0.459	611.42	0.338	34.57
LLM `Claude 3.5 Sonnet`	Billions	**0.750**	**0.895**	**0.640**	337.80	0.389	39.22
Token Classification `roberta-base`	125M	0.690	0.838	0.574	**1.23**	**0.441**	**0.10**

Table 2: Summary of the results. Evaluation times are reported in minutes. For experiments relying on external APIs (`text-embedding-004` and `Claude 3.5 Sonnet`), times might vary depending on rate limits, connection speeds, etc.

3.1 Downstream Task – Paraphrase Detection

Model	STS	STS + DSD
`all-MiniLM-L6-v2`	0.720	**0.808**
`all-mpnet-base-v2`	0.795	**0.868**

Table 3: Comparison of accuracies on the PAWS-Wiki Labeled using uniquely STS or combining STS with DSD.

DSD improves the performance on the task with no fine-tuning.

2. Experimental Setup

We propose **5 methods** to tackle the task of DSD:

SHAP-DSD* — Based on the SHAP framework [1]. We calculate the dissimilarity score as:

$$\mathrm{DissimilarityScore}(a, b) = 1 - \mathrm{CosineSimilarity}(a, b)$$

Tokens with dissimilarity above a threshold are considered as dissimilar.

LIME-DSD* — Similar to SHAP-DSD, but with the explanation weights predicted by LIME [2].

LLM-DSD — Uses an LLM to detect the dissimilar spans.

Token-Classification-DSD — A model is fine-tuned to predict dissimilar spans.

Embedding-DSD — This method constitutes a **novel contribution**.

We calculate all possible unigram, bigram, ..., *n*-gram replacements from the first sentence to the second. For example, for the following 2 sentences:

```
the bird flies fast over the hill
the car rides fast over the hill
```

If we were considering the trigram `['the', 'bird', 'flies']`, we would get the replacements:

```
1 the bird flies fast over the hill
2 the the bird flies over the hill
3 the car the bird flies the hill
4 the car rides the bird flies hill
5 the car rides fast the bird flies
```

We then calculate the *similarity gain* for each *n*-gram by comparing the resulting cosine similarity to the *base similarity* (i.e., the cosine similarity between the original sentences). Each unigram is assigned the gains coming from each of the *n*-grams in which it appears. These are then combined through an *aggregation function*:

$$\mathrm{AggrGain}_{\mathrm{unigram}} = \frac{1}{n} \cdot \sum_{i=1}^{n} \frac{\mathrm{gains}_i}{i}$$

Unigrams with a gain above a threshold are considered as dissimilar.

* Methods that require a threshold. We find the optimal threshold by evaluating on the validation split.

Want to know more?

dmlls.github.io/dissimilar-span-detection

References:

[1] Marco Tulio Ribeiro et al. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135-1144.

[2] Scott M Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765-4774.

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